



Yr8 COMPUTER SCIENCE

Revision Booklet

Assessment Point 4

You could have questions based on any of the lessons you have had so far this year plus your lessons from year 7. This revision guide includes all the key knowledge you need to know in preparation for your first Computing Assessment of Year 8.

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Glossary

Algorithm - A set of clear steps to solve a problem or complete a task.

Animation - Movement added to slides to make things appear or stand out.

Binary - A way computers count using only 0 and 1.

Bit - The smallest piece of data in a computer, just a 0 or 1.

Byte - A group of 8 bits.

CPU (Central Processing Unit) - The “brain” of the computer that runs programs and processes data.

Data - Information that a computer uses or stores.

Denary (Decimal) - The number system humans use (0–9).

Hard Drive - Where files and programs are stored long term.

Input Device - Something you use to put data into a computer (e.g. keyboard, mouse).

Output Device - Something that shows data from a computer (e.g. screen, printer).

Iteration - Repeating steps again and again until something is finished.

Layout - How things are arranged on a page or slide.

Malware - Bad software that can harm your device or steal your data.

Memory (RAM) - Stores data and programs while the computer is using them.

Motherboard - The main board that connects all parts of a computer together.

Power Supply (PSU) - Gives power to the computer.

Presentation - A set of slides used to show information to others.

Processor - Another name for the CPU.

Program - A set of instructions that tells a computer what to do.

RAM - Short-term memory that holds things you are currently using.

Selection - Making a choice in a program (e.g. if something happens, do this).

Sequence - The order that instructions happen in.

7.1 School Systems & Software

Lesson 2 - Microsoft Outlook (Emails)

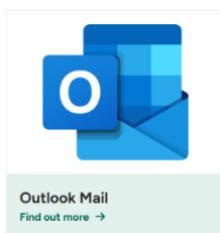
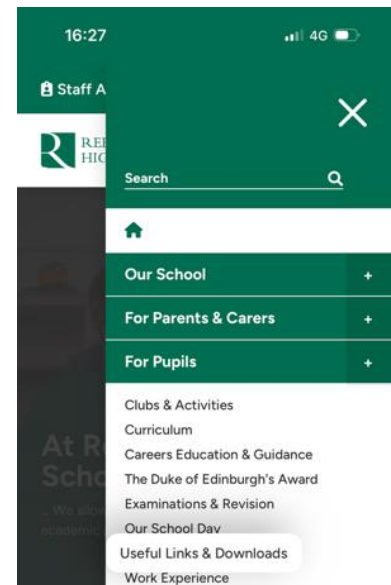
To access your emails from **home**:

1. Go to the school website www.reephamhigh.org.uk
- 2.

On a **PC**, hover over “For Pupils” and select “Useful Links & Downloads”.

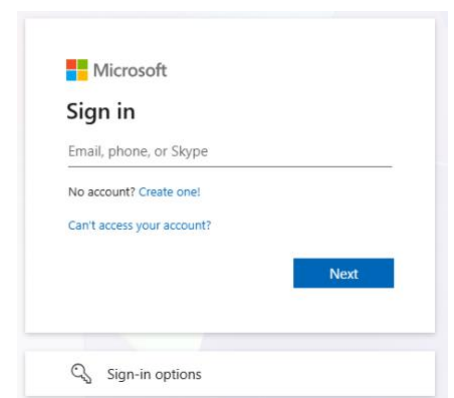
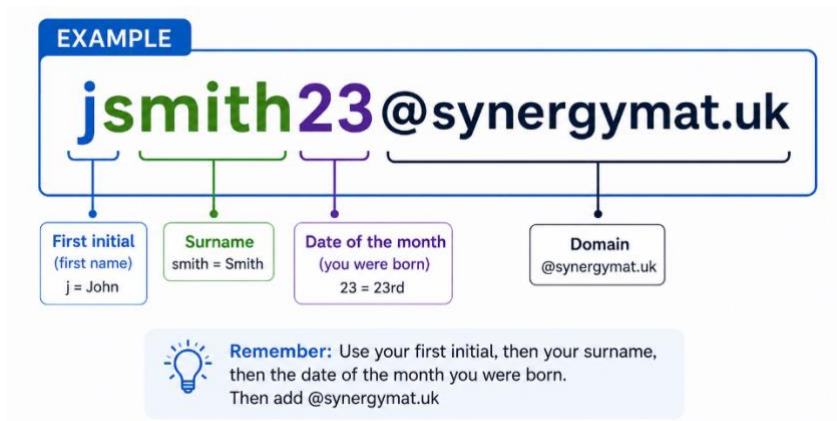


On a **phone or tablet**, click on the three lines, click the plus next to “For Pupils” and then select “Useful Links & Downloads”.



3. Scroll down and click on the Outlook Mail icon.

4. Sign in with your school email and password (you set your password up in your first Computing lesson).



7.2 Computer Systems

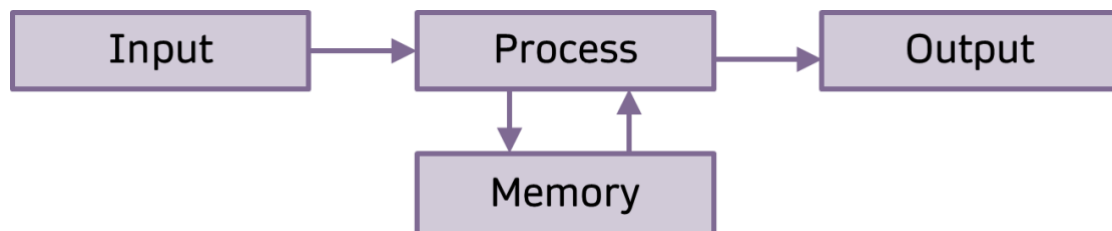
Lesson 2 – Designing Presentations

When designing an **effective presentation** we must **consider the following**:

- Colour - When designing a presentation, **choose colours carefully** so that your text is easy to read and important information stands out.
- Layout - Think about the **layout** of each slide by arranging text and images neatly so your audience can follow the information easily.
- Size - Make sure the **size** of your text and images is appropriate so that everyone in the room can see them clearly.
- Animation - Use **animations only when they improve your presentation**, such as emphasising important points instead of distracting your audience.
- Simplicity - Keep your slides **simple** by including only the most important information and avoiding too much text or too many images.
- Consistency - Be **consistent** by using the same fonts, colours, and styles throughout your presentation to give it a professional appearance.

Lesson 3 - Input and Output Devices

All computer **systems** do these four things:



Data is entered into the computer through **input devices**, and the computer presents the output through **output devices**:

Input Devices	Output Devices
Allows the user to put data into a computer system.	A computer will send data out of these devices.
Examples: • Mouse • Keyboard • Scanner • Camera • Microphone	Examples: • Monitor (Screen) • Printer • Speakers • Headphones • Projector

Lesson 4 - Computer Components

All computer systems have the same parts. But what are they?

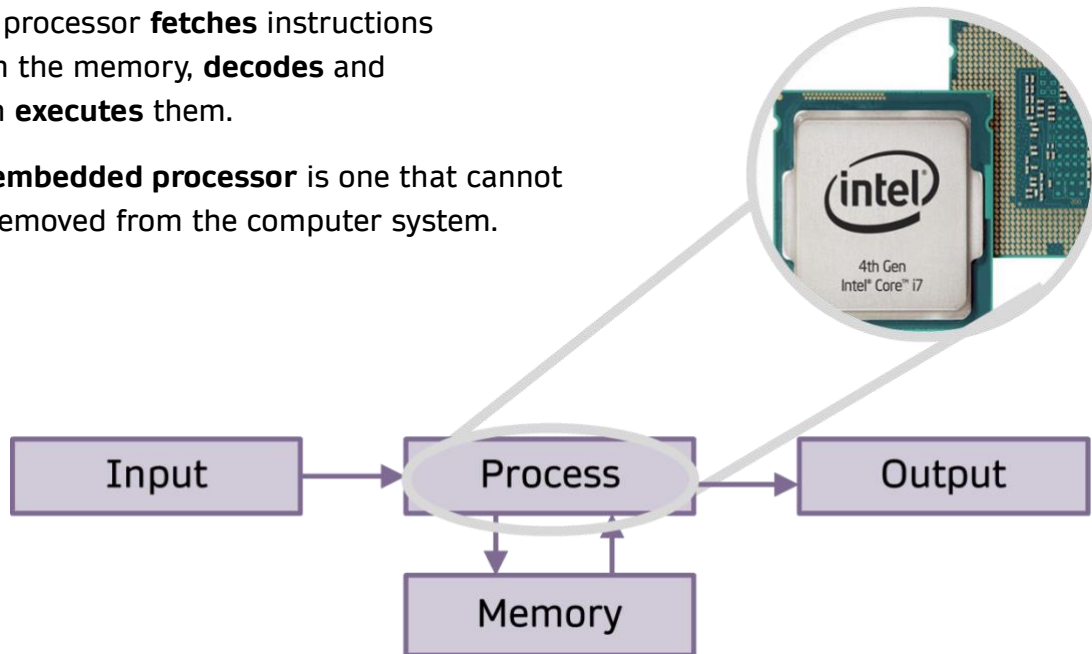
<u>Motherboard</u>	A circuit board that connects all the other parts of the computer together.	
<u>CPU / Processor</u>	Central Processing Unit is the brains of the computer and processes all the data.	
<u>RAM</u>	Random Access Memory - Stores the data and programs that you are currently running.	
<u>Graphics and Sound Cards</u>	Allow you to see and hear what is on the computer clearly.	
<u>Hard Drive</u>	Stores your programs and files.	
<u>Power Source/PSU</u>	To give your computer power attached to the wall plug.	

Lesson 5 - CPU

Processing is done by the processor aka **Central Processing Unit (CPU)**.

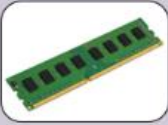
The processor **fetches** instructions from the memory, **decodes** and then **executes** them.

An **embedded processor** is one that cannot be removed from the computer system.



Lesson 6 - Memory and Storage

Both memory and storage devices store data on our devices however they do it for different reasons:

Memory	Storage
Can directly access the CPU.	Must transfer all instructions to the memory before they can be executed.
Hold data currently being used.	Stores all programs and files not in use .
Short Term – Volatile	Permanent – Non-Volatile
 RAM  Cache	 Memory Stick  SD  Hard Drive  Solid State Drive  DVD  CLOUD STORAGE

Lesson 7 - Binary

How computers store count:

Computer use millions of electronic circuits and switches which can either be On or Off. The computer will represent these are binary digits which are either 1 or 0.

0 or a 1 = 1 bit (Binary Digit)

4 Bits = 1 Nibble

8 Bits 1 Byte

How humans count:

We count using the numbers 0-9, this is called **Denary** or sometimes Decimal.

How to convert binary to denary:

Binary = 0 1 0 0 1 0 1 1

Step 1 – Put binary number into this place value table:

Place Value	128	64	32	16	8	4	2	1
Binary	0	1	0	0	1	0	1	1

Step 2 – If there is a 1 in a column, add the column heading together.

For example:

$$\begin{array}{r} 64 \\ + \quad 8 \\ + \quad 2 \\ + \quad 1 \\ \hline \text{Denary} = \quad 75 \end{array}$$

How to convert denary to binary:

Denary = 131

Step 1 – Create your place value table:

Place Value	128	64	32	16	8	4	2	1
Binary								

Step 2 – Put a number 1 in the place values that can be added together to create the denary number. For example: $131 = 128 + 2 + 1$

Place Value	128	64	32	16	8	4	2	1
Binary	1						1	1

Step 3 – Fill any gaps with 0s.

Place Value	128	64	32	16	8	4	2	1
Binary	1	0	0	0	0	0	1	1

131 in binary is 10000011

7.3 Problem Solving

Lesson 2 - Sequences

What is an algorithm?

A set of step-by-step instructions to solve a problem or complete a task.

What is a sequence in Computing?

It's the exact order in which steps happen – changing the order can change the result.

Example of a sequence in everyday life:

Brushing your teeth: first get your toothbrush, then put toothpaste on it, then brush.

7.3.7 Selection in coding using Small Basic Turtle

Lesson 3 - Selection

What is selection in Computing?

To make a choice which steps should be followed in an algorithm

Example of selection in everyday life:

If it's raining, you put on a raincoat.

If it's sunny, you grab your sunglasses.

You make a choice based on a condition – code uses an if statement!

Lesson 4 - Iteration

What is iteration in Computing?

Iteration means which instructions we repeat when solving a problem.

Example of iteration in everyday life:

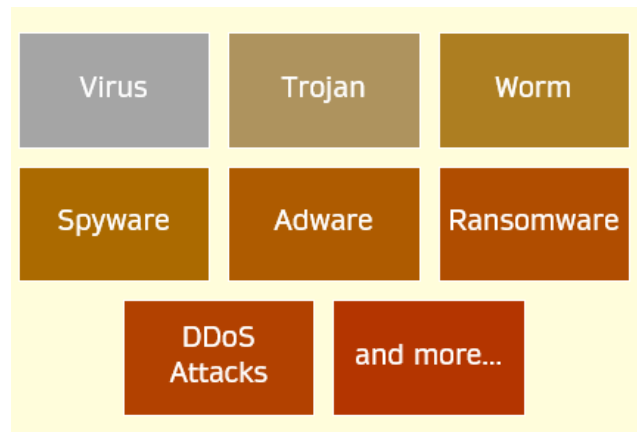
You repeat the same motion, brush up then brush down, over and over until the job is done.

7.4 Navigating the Digital World

Lesson 2 - Malware

Malware (malicious software) is designed to harm your device or steal your personal data.

Here are some examples of malware:



Some signs you might see if your device was infected with malware may be:

1. Unable to use your device properly.
e.g.
 - Device running very slowly.
 - Unwanted pop-ups and ads appearing frequently.
 - Your device crashing or freezing unexpectedly.
2. Personal data being stolen and used by criminals.
3. Loss of files and programs

Some ways to protect your device:

1. **Think before you click** – don't open links or attachments from people you don't know.
2. **Use strong passwords** – long, hard-to-guess, and never shared with friends.
3. **Keep your software updated** – updates often fix security holes.
4. **Install antivirus software** – helps spot and block bad stuff.
5. **Tell an adult** – if something seems strange or scary, don't deal with it alone.

Lesson 5 - Computer Legislation

Computer Misuse Act

The **Computer Misuse Act** is a law that protects computers and data from illegal access or misuse. It covers actions like hacking, spreading viruses, and accessing data without permission.

How could someone break the law under the CMA?

Someone could break the CMA by hacking into a computer, stealing or changing data, or spreading viruses that damage computers or networks.

Data Protection Act

The **Data Protection Act** is a law that helps protect personal data. It requires organizations to store and handle data in a way that keeps it safe and respects people's privacy.

How could someone break the law under the DPA?

Someone could break the DPA by sharing personal information without permission or not keeping data secure, like leaving it on an unprotected computer.

Copyright, Designs and Patents Act

The **Copyright, Designs and Patents Act** protects the rights of creators and inventors. It stops people from copying, using, or distributing someone else's work without permission.

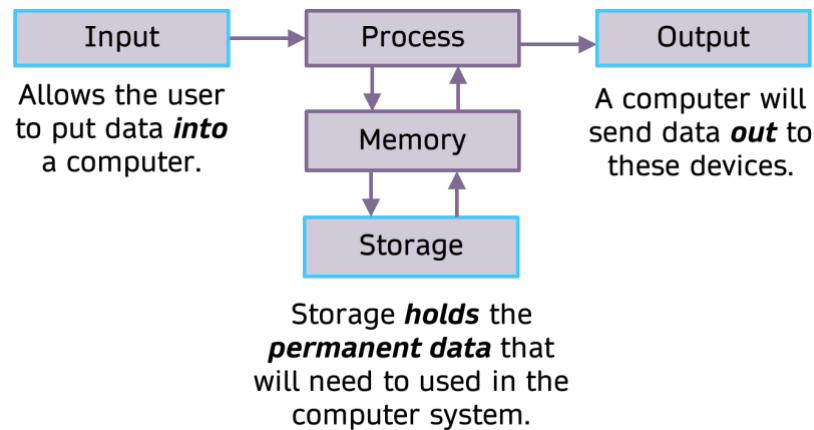
How could someone break the law under the CDPA?

Someone could break the CDPA by downloading or sharing someone's work without their permission, like copying a song or using software without paying for it.

8.1 Computer Systems

Lesson 1 - Computer Systems

All computer systems do these four things.



Computer systems have both input and output devices. See example below:

Input Devices	Output Devices
Microphone	Speakers
Keyboard	Headphones
Mouse	Monitor
Checkout scanner	Projector
Camera	Printer
Motion Sensor	

Remember that some examples can be both input and output:

- Headsets
- Touchscreens
- Games controller if it allows haptic feedback (vibrations)

Typical Questions

In your tests, you will see the same types of questions again and again.
If you practise these, you'll feel much more confident!

1. Quick Knowledge Questions (1 mark)

These test what you **remember**.

You might be asked to:

- State what something means
 - Name a component
 - Define a key word
-

2. Give Examples (1-2 marks)

You need to **list examples**.

You might be asked to:

- Give input or output devices
 - Give ways to stay safe
 - Give features of a good presentation
-

3. Describe Questions (1-2 marks)

You explain **what is happening**.

You might be asked to:

- Describe what code does
 - Describe a sign of malware
 - Describe a computer process
-

✅ 4. Explain Questions (2–3 marks)

You explain **how or why something works**.

You might be asked to:

- Explain the difference between two things
- Explain how something works
- Explain why something happens

👉 Tip: Use connectives like *because*, *so*, *this means*.

✅ 5. Step-by-Step Questions

You show **processes in order**.

You might be asked to:

- Explain the CPU cycle (fetch, decode, execute)
- Explain steps in logging in
- Describe an algorithm

👉 Tip: Put answers in the correct order.

✅ 6. Fill in the Gaps / Tables

You complete missing information.

You might be asked to:

- Complete tables
- Fill in keywords
- Match terms to definitions

👉 Tip: Read carefully for clues.

✅ 7. Tick Box Questions

You choose the **correct answer**.

👉 Tip: Read all answers before picking one!

✅ 8. Code and Real-Life Questions

You apply your knowledge.

You might be asked to:

- Describe what code is doing
- Explain a real-life computing situation

👉 Tip: Link your answer to what you already know.

✅ 9. Binary and Data Questions

You work out simple calculations.

You might be asked to:

- Convert binary to denary
- Show binary shifts

👉 Tip: Make sure you show your working out!

✅ 10. Compare Questions (2 marks)

You explain **the difference between two things**.

Example topics:

- RAM vs ROM
- Memory vs storage
- VR vs AR

👉 Tip: Use words like *whereas* or *however*

★ Top Exam Tips

- Read the command word carefully (e.g. *describe*, *explain*)
- Use full sentences for longer answers
- Give examples when you can
- Check how many marks the question is worth

Command Words

Tick

Select the correct answer from a list (usually a box).

Identify

Pick out or name the correct answer.

Give

Provide an example or short answer.

Name

Write the correct word or item.

State

Write a short, clear answer (usually a fact, no explanation needed).

Show

Work out an answer and display your working.

Complete

Fill in the missing information.

Define

Give the exact meaning of a word or term.

Describe

Say what something is like or what is happening.

Explain

Give clear reasons or describe how and why something happens.

Modelled Questions

Part A – 7.1 Introduction to Computing

1 When selecting a password, it is important that it is not easy to guess.

Tick (✓) one box to identify which would be the strongest password.

[1]

☐

Football123

☐

LetMeIn!

☐

TreeHouse55

☒

W@terM3lon 92! ✓

2 Identify two additional elements of a strong password.

The first has been written for you.

[2]

1 Special Characters

2 Capital Letters ✓

3 Numbers ✓

3 Complete this student's email address by filling in the missing word.

[1]

sharrington18@ synergy mat.uk ✓

4 Explain the purpose of the **subject** in an email.

[1]

The person receiving the email will know

what it is about before opening it. ✓

5 Describe a situation where a **carbon copy (CC)** might be used when sending an email. ①

A teacher emails me homework and sends a
copy to my parents, but they don't need to
take action.

6 Write an email to your teacher to ask them for help with your homework. Consider how professionally it is written and include all the essential elements of an email. ④

Subject:

Homework Help / subject

Email:

To Mrs Peebles, / greeting

I am struggling to access my
wayground homework as I was absent
when we signed up. / content

Could I have some help with this?

many Thanks,
molly / sign off

7 Explain how to log into Microsoft Teams from home, if you do not have Teams installed on your device. (3)

Go to the Reepham High School website,
click on "For Pupils" and then "Useful
Links". You then click on the Teams icon and
log in with your username and password.

8 The table contains applications (apps) you have used this year and their purpose.
Complete the table by either naming the app or it's purpose. (3)

App	Purpose
Outlook	Used to send and reply to emails about schoolwork.
Microsoft Teams	Get work for lessons.
One Drive	A cloud storage service for saving and sharing files online.

9 Below are three scenarios that someone might experience online.

For each scenario, **describe two** pieces of advice you would give to ensure they stay safe while using the internet.

Scenario 1: Receiving a Friend Request from a Stranger

(2)

They should not accept the friend request
and block them if they try to get in contact
again.

Scenario 2: Being Pressured to Share Personal Information

(2)

Remind them to keep their personal
information private and to share their
concerns with a trusted adult.

Scenario 3: Experiencing Cyberbullying

(2)

Block and report the bully and report
to a trusted adult.

Part B – 7.2 Computer Systems

10 Keyboards and Mice are examples of input devices.

Give two examples of output devices.

(2)

1 Printer ✓

2 Speaker ✓

11 Which computer component connects all the parts of the computer together?

Tick (✓) one box to identify the correct answer.

(1)

☐

RAM

☐

Hard Drive

☒

Motherboard

☐

CPU

12 Describe the three-step cycle that the CPU completes every time an instruction is processed.

(3)

Instructions are fetched from memory,
decoded so they are understood and then
executed (carried out).

13 The table contains a description of the terms **Memory** and **Storage**.

Complete the table by describing the purpose of memory.

(1)

Term	Memory	Storage
Purpose	Stores currently used data and files ✓	Store files and programs not currently in use.

14 **Complete** the table by converting the binary numbers into denary and the denary number into binary.

(3)

Binary	Denary
1010	10 ✓
1100 ✓	12
11001100	204 ✓

15 **Identify two** additional examples of characters.
The first has been written for you.

(2)

- 1 Symbols
- 2 Numbers ✓
- 3 Letters ✓ 800 ← This means "benefit of the doubt" its close enough to the correct answer.

Retrieval Practice Questions

1. What is an algorithm?
2. What is a sequence in computing?
3. What is selection in computing?
4. What is iteration?
5. What does CPU stand for?
6. What is the job of the CPU?
7. What does RAM store?
8. What is a hard drive used for?
9. What is an input device?
10. Give one example of an input device.
11. What is an output device?
12. Give one example of an output device.
13. What is binary?
14. What numbers does binary use?
15. How many bits are in a byte?
16. What is malware?
17. Give one sign of malware on a device.
18. What is the motherboard?
19. What should you think about when designing a presentation?
20. What is the Computer Misuse Act?

Retrieval Answers

1. A set of step-by-step instructions to solve a problem.
2. The order in which instructions happen.
3. Making a choice in a program (for example, using if statements).
4. Repeating steps again and again.
5. Central Processing Unit.
6. To process data and run instructions (the brain of the computer).
7. Data and programs currently being used.
8. To store files and programs long term.
9. A device used to put data into a computer.
10. Keyboard, mouse, or microphone.
11. A device that shows data from a computer.
12. Monitor, printer, or speakers.
13. A number system used by computers.
14. 0 and 1.
15. 8 bits.
16. Harmful software that damages devices or steals data.
17. Slow device, pop-ups, or crashes.
18. The main board that connects all computer parts together.
19. Colour, layout, size, animation, simplicity, or consistency.
20. A law that protects computers and data from misuse